

ROHIT PILLAI

✉ Kurralta Park, Adelaide

SUMMARY — Data Engineer with strong experience in data modelling and building end-to-end data pipelines on Databricks and Azure, and a growing focus on machine learning solutions. I bring extensive exposure to financial data within the construction industry, delivering reliable analytics that support operational and leadership decision making.

TECHNICAL COMPETENCIES

Data Platforms: Declarative Pipelines, PySpark, Jobs, Databricks Apps, Microsoft Fabric (Power BI, Azure data factory)

Data Governance: Unity Catalog, Schema privileges, Configuring Power BI audience and enabling row-level security

Reporting: Power BI Service, DAX, PowerQuery

Databases: MySQL, MongoDB

Workflow & Streaming: GCP Pub/Sub (Streaming Data)

Programming Languages: Python, SQL, R

Version Control & Documentation: GitHub, Markdown, SQLDBM, ChartDB

WORK EXPERIENCE

Data Engineer

July 2025 – present

John Holland

Assigned to: Torrens to Darlington Project (T2D SA)

- Built PySpark and Python-based validation frameworks with a Streamlit UI to replicate 25 compliance checks performed by external auditors using their proprietary Compliance IQ tool. Validated ledger data prior to client submission, reducing audit risk and cutting manual effort from 3 people over 3 days to 3 hours.
- Developed an end-to-end analytical solution for the finance team to track a critical project KPI—vendor payments against SLA. Designed data pipelines and orchestrated jobs to maintain a materialized view, which powered a live Power BI dashboard using DirectQuery, enabling leadership decision-making and driving a 3–4% month-over-month improvement in SLA compliance.
- Created ER diagrams to translate the business problem of managing data from different sources and Excel files to solve a manual reporting problem. Further created the logical layer by splitting the flat format into line items, modelling the foreign and primary keys, and implementing the physical layer in Databricks.
- Built automated bi-weekly management reports using Power Automate. Delivered walkthrough sessions across finance and commercial teams to demonstrate dashboard capabilities, gather feedback, and iteratively improve reporting based on user needs.
- Managed data governance through Unity Catalog by restricting catalog and schema privileges for the T2D workspace. Further ensured governance for Power BI dashboards through Power BI apps, audiences, and row-level security.
- Worked closely with finance stakeholders to understand construction and finance domain processes, translating business requirements into reliable data analysis.

Data Analyst

Jan 2025 – June 2025

Get Sliced

Contract project: Database Design and Analytics

- Designed the entire database for a local food business using an open-source database solution to connect to POS applications and inventory management systems.
- Created ER diagrams to understand business pain points, leaving room for future scalability, and developed the corresponding database schema designs.
- Developed test cases and conducted monitoring to ensure data quality standards for the ingested data.
- Developed a Power BI dashboard to visualise critical sales trends across different items, enabling efficient inventory management, cost savings, and informed promotional decisions.

Capgemini Technology Services

Sept 2021 – Jan 2023

Graduate Data Engineer

Client: Independence Blue Cross (Health Insurance Provider, US)

- Collaborated on migrating an on-premise database to the cloud using the Azure platform, with a focus on transformations implemented in Azure Data Factory.
- Developed an in-house central repository for different teams to access APIs along with their metadata and documentation, creating a one-stop platform to inspect APIs and request access from owners.
- Designed process maps using Visio and delivered stakeholder-friendly presentations using PowerPoint to communicate architecture to non-technical client managers.
- Supported cross-functional collaboration by resolving integration queries in Azure Data Factory with the API team, and mentored junior team members during project ramp-up.

PROJECTS

Database Migration: On-Prem to Azure Cloud

Oct 2024

Tech Stack: Microsoft SQL Server, Azure Key Vault, Azure Data Factory

[View on GitHub](#)

- Developed and delivered a data migration pipeline to transfer the AdventureWorksLT database from on-premises SQL Server to Azure using a self-hosted integration runtime and dynamic SQL/JSON.
- Structured the migration workflow using Azure Data Lake Gen2 (bronze, silver, gold layers), ensuring data integrity.
- Used GitHub markdown for maintaining clear documentation and reproducibility.

Customer Churn Prediction – Australian Health Insurers

Oct 2024

Tech Stack: Python, Excel, MySQL

- Conducted exploratory analysis on 20,000+ rows of APRA-sourced data to uncover business insights and identify key churn indicators.
- Created correlation heatmaps and scatter plots to support feature selection and improve model interpretability.
- Preprocessed data using outlier removal, one-hot encoding, and cross-validation; trained an enhanced KNN model to classify churn behaviour.
- Delivered insights on policyholder dissatisfaction, prompting recommendations for better exit surveys and targeted retention strategies.

Sales Analysis for Hardware Store

Sept 2024

Tech Stack: Power BI, MySQL, Excel

[View on portfolio website](#)

- Developed and maintained a unified Power BI dashboard integrating sales, marketing, product, and finance data.
- Cleaned, transformed, and modelled data using SQL and Excel to support key business performance indicators.
- Visualised trends via DAX measures and custom visuals, improving clarity and enabling decision-making for store managers.

Forecasting Household Energy Usage

Jul 2023

Tech Stack: Python, Jupyter Notebooks, Scikit-learn

[View on portfolio website](#)

- Analysed seasonal energy consumption patterns and identified key influencing factors via feature correlation heatmaps and box plots.
- Built predictive models using Linear Regression and Support Vector Regression, comparing model accuracy (MAE) across use cases.
- Delivered results through clear data storytelling, using plots and markdown reports to support stakeholder understanding.

EDUCATION

The University of Adelaide, Adelaide

Jan 2023 - May 2025

Master of Data Science

Savitribai Phule Pune University, Pune

Sept 2017 - Aug 2021

Bachelor of Engineering in Computer Science

CERTIFICATION

Databricks Certified Data Engineer Associate

March 2026

Microsoft Certified Azure Data Fundamentals (DP-900)

Oct 2024

Academy Accreditation- Databricks Lakehouse Fundamentals

Oct 2024

EXTRACURRICULAR

Robocon India

Jan 2019

Team lead of coding department predominantly on C++ Arduino Uno